ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

ROBOTICS AND INDUSTRIAL AUTOMATION

PAPER 2 · PRACTICE D

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 4. Practical paper in the style of the DHET second papers: design the breadboard circuit and flowchart of a parcel-weighing station with a load cell, an LCD and a stepper-motor gate, program it on an Arduino, then draw the layout of the guarded cell, write a PLC ladder program for its conveyor and complete a risk assessment.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 4. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of THREE questions. Answer ALL the questions.
2. This is a practical paper. Use Fritzing for circuits, draw.io for flowcharts, layouts and ladder diagrams, and the Arduino IDE for programs. If you do not have a computer, draw and write your answers neatly by hand.
3. Put the question number and your name or examination number on every printout or page. Save your work regularly.
4. Programs are not marked strictly on syntax.

QUESTION 1

A courier depot wants a prototype station that weighs parcels and sorts them into price bands. The operator places a parcel on a scale (a load cell with an HX711 amplifier) and presses the WEIGH button. The station shows the mass and the band on an LCD: up to 2 kg is band A, more than 2 kg up to 5 kg is band B, and more than 5 kg is too heavy. For bands A and B a stepper motor opens a gate so that the parcel can be pushed through. A parcel that is too heavy is rejected: a red LED comes on and a buzzer beeps three times.

Components:

- Arduino Uno
- Load cell with an HX711 amplifier module
- 16 × 2 LCD with an I2C module
- 28BYJ-48 stepper motor with a ULN2003 driver board
- Push button (WEIGH)
- Red LED and green LED, each with a 220 Ω resistor
- Active buzzer
- Breadboard and jumper wires

1.1 Using the components above, design the breadboard circuit of the weighing station in Fritzing. (14)

1.2 Draw a flowchart of the logic of the weighing station, including reading the mass, choosing the band, opening and closing the gate, and rejecting a parcel that is too heavy. (14)

1.3 Save and print (or neatly present) the circuit and the flowchart. (2)

[30]

QUESTION 2

Use these connections: WEIGH button to pin 2 (INPUT_PULLUP, so it reads LOW when pressed); HX711 DT to pin 3 and SCK to pin 4; buzzer to pin 6; red LED to pin 7; green LED to pin 12; ULN2003 IN1–IN4 to pins 8–11; LCD SDA to A4 and SCL to A5. The stepper motor needs 2 048 steps for one full turn.

2.1 Write an Arduino program for the weighing station that does the following: (28)

• Sets up all the pins, the scale (set the calibration factor and tare it), the LCD and the serial monitor, and shows "Place parcel" on the LCD.

• When WEIGH is pressed, reads the mass in kilograms (average of 10 readings) and shows it on line 1 of the LCD, for example "Mass: 3.25 kg".

• Shows "No parcel" and does nothing else if the mass is less than 0,05 kg.

• Shows "Band A" or "Band B" on line 2, switches the green LED on, opens the gate by turning the stepper 90°, waits 3 seconds, closes the gate and switches the green LED off.

• For a parcel heavier than 5 kg, shows "Too heavy", switches the red LED on and beeps the buzzer three times, then switches the red LED off.

• Counts the parcels in band A, band B and the rejected parcels, and prints the three counters on the serial monitor after every parcel.

• Is well commented.

2.2 Save and print (or neatly present) the program. (2)

[30]

QUESTION 3

The depot will build a full-size version in which a conveyor carries the weighed parcels to a robot that stacks them. The conveyor is controlled by a PLC.

3.1 Draw a labelled top view (layout) of the full-size cell. Show the weighing station, the conveyor, the robot and its work envelope, the stacking area, the safety fence with an interlocked gate, a light curtain where parcels enter, the emergency stop buttons and the operator's position. (12)

3.2 Draw a PLC ladder program for the conveyor with these inputs and outputs: START (NO) I0.0, STOP (NC) I0.1, emergency stop (NC) I0.2, light curtain clear (NC contact, opens when the beam is broken) I0.3, conveyor motor Q0.0, green "running" lamp Q0.1 and red "stopped" lamp Q0.2. START must latch the conveyor; STOP, the emergency stop or a broken light curtain must stop it, and START must be pressed again to restart. (15)

3.3 Copy and complete the risk assessment table below for THREE hazards in the full-size cell. (10)

Hazard	Possible injury or damage	Risk (low / medium / high)	Control measure
1.			
2.			
3.			

3.4 Save and print (or neatly present) your layout, ladder program and risk assessment. (3)

[40]

TOTAL: 100

END OF QUESTION PAPER